systems. Together, these viewpoints offer a comprehensive understanding of how the geometry of attention influences the transformer behavior, as we elaborate in sections 4.1 and 5 with further technical details and empirical observations.

**Geometry of attention and training transfromers.** Transformer models are typically over-parameterized (Fan et al., 2019; Liu et al., 2021a; Wang & Tu, 2020), which leads to a non-convex optimization landscape (Bassily et al., 2018; Choromanska et al., 2015; Liu et al., 2022). In such cases, fast convergence of the loss minimization relies on the Polyak-Łojasiewicz (PL) condition (Bassily et al., 2018; Polyak, 1963; Liu et al., 2022). We demonstrate that the probability of satisfying this condition is connected to the attention graph's eigenvalues, which are in turn linked to its ORC. This relationship establishes ORC as a key proxy for understanding convergence properties in transformer optimization. This is implied from Lemma 4.2, which states that a larger smallest eigenvalue in the attention map corresponds to faster convergence of gradient descent on the transformer. Meanwhile, a monotonic function of the Ricci curvature controls the minimum eigenvalue of the attention map. It follows that the ORC is linked to the convergence rate in transformers training –a link that we empirically explore in Section 5. In Section 4.1 we elaborate on the theoretical steps leading to this finding, and the full proof and assumptions are included in Appendix A. This result introduces a novel geometric perspective on loss optimization in transformers, offering new avenues for improving their training processes.

**Graph geometry, statistical mechanics, and robustness of transformers.** Extending our geometric analysis beyond training dynamics and adapting a physics perspective, we interpret the attention graph as a system of interacting particles, where tokens represent particles and attention scores quantify inter-particle interaction strengths. In particular, the output of a deep transformer could be formulated as the solution to the continuous-time approximation of attention dynamics through a forward pass, where each attention layer is modeled as a time snapshot of particle interactions. The dynamics of this particle system, detailed in Appendix B, reveal how token representations evolve through attention layers. This analogy allows us to apply principles from statistical mechanics and dynamical systems to transformers, inheriting properties of networked systems that propagate signals across interconnected units. A crucial property of such networked systems is their robustness to signal perturbation, which for transformers translates to maintaining performance on noisy inputs. This robustness is linked to the system entropy and, consequently, to Ricci curvature (Demetrius & Manke, 2005; Pouryahya et al., 2017), providing a theoretical basis for using curvature to assess and enhance transformer robustness. As the model processes inputs through a forward pass, we observe shifts in the ORC distribution, trending toward more positive values or reducing the frequency of smaller curvature values. This observation is in line with previous findings on emergent clustering in attention layers through a forward pass (Geshkovski et al., 2023) (see Appendix B), due to the connection between ORC and weight clustering, which we explain in Appendix C. We discuss the details of how the attention dynamics leads to the shifts in the ORC distribution in Appendix B, where we also show this phenomenon on additional transformer models. These shifts, visualized in Figure 3, suggest a strong connection between attention mechanism dynamics and the geometric properties that influence transformer robustness. Thus, Ricci curvature emerges as a multifaceted concept in transformer analysis: it informs training dynamics, serves as a geometric signature of robustness and performance, and offers a promising tool for model improvement. Our empirical results in Section 5 further elucidate these diverse roles and their practical implications.

## 4 Ricci curvature and training of transformers

In this section, we explore in detail the above-mentioned theoretical and practical implications of Ricci curvature in the training of transformers. We first establish the mathematical foundations linking the eigenvalue spectrum of the attention graph to the optimization process in over-parameterized models, which allows us to demonstrate how Ricci curvature influences gradient descent in transformers. We then propose a practical method to manipulate curvature distributions during training, which not only enables us to empirically validate our theoretical findings but also offers an approach to train enhanced transformers, following our findings on the role of curvature in their learning dynamics and robustness.

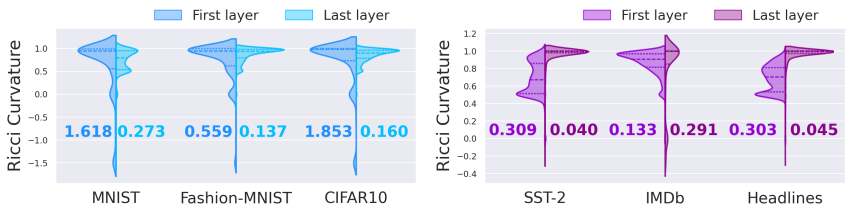


Figure 3: The ORC distributions of the first and last attention layers of ViT (blue) for a batch of MNIST, Fashion-MNIST, and CIFAR10 images, and BERT-Tiny (purple) for a batch of SST-2, IMDb, and Headlines texts. The curvature distribution shifts towards more positive values through a forward pass; quantified by the Wasserstein distances of the ORC distributions from a Dirac mass at the ORC value of 1, annotated next to each distribution (smaller means more positive leaning).

### 4.1 Attention graph, Ricci curvature, and gradient descent

In Section 3 we explained our main theoretical findings on the connection between the Ricci curvature of attention and gradient descent-based training of transformer models. Here we elaborate on the steps leading to this result, which rely on a connection we establish between the eigenvalue spectrum of transformers and the optimization of over-parameterized models. Following standard practices in the theoretical analysis of transformers (see, e.g., (Dovonon et al., 2024; Von Oswald et al., 2023; Abbe et al., 2024)), we derive our theoretical results in an idealized setting with a set of simplifying assumptions, stated here and in Appendix A. Our empirical investigation further demonstrates that the key implications of our theoretical findings extend beyond these idealized conditions, proving relevant and applicable to common transformer models in practice.

**Eigenvalues of transformers.** Consider a single-head transformer of depth $L$, with each block containing an attention layer, a skip-connection, and a projection. For this analysis, we assume the feed-forward layer in the attention block is a projection by $W_{p,l}$. Hence, the output of each layer is

$$X_{l+1} = X_l + A_l X_l W_{V,l} W_{p,l}, \tag{4}$$

where $W_{V,l}$ is the value weight matrix defined in Section 2, for block $l$. Following Dovonon et al. (2024), we further assume that all attention blocks are identical and remove the layer indexing. Stacking the column to obtain a vectorized input, $\mathbf{x}$, we can write

$$F(\mathbf{x}) = \mathbf{x}_L = (I + H \otimes A)^L \mathbf{x}, \tag{5}$$

$$\lambda^F{}_{ij} = \left(1 + \lambda^A{}_i \lambda^H{}_j\right)^L, \tag{6}$$

where $F(\cdot)$ and $\lambda^F{}_{ij}$ denote the transformer and its eigenvalues, $H := W_p{}^T W_V{}^T$, and $\lambda^A{}_i$ and $\lambda^H{}_j$ are the eigenvalues of $A$ and $H$. Note that relaxing the above-mentioned assumptions is rather straightforward, though distracting and not amenable to the here provided analysis.

**Tangent kernel and convergence of gradient descent.** Transformers are often over-parameterized (Fan et al., 2019; Liu et al., 2021a; Wang & Tu, 2020) and hence essentially non-convex (Bassily et al., 2018; Choromanska et al., 2015). The Polyak-Łojasiewicz (PL) condition (Bassily et al., 2018; Polyak, 1963) implies exponential convergence of gradient descent in non-convex regimes (Liu et al., 2022). Given a model $f_\theta(\cdot)$ parameterized by $\theta$, we say the loss function $\mathcal{L}(\theta)$ is $\mu$-PL if $\|\nabla\mathcal{L}(\theta)\|^2 \geq \mu L(\theta)$. Let $K_\theta := (\nabla f_\theta)(\nabla f_\theta)^T$ and $\lambda_1(K_\theta)$ be the tangent kernel of $f$ and its minimum eigenvalue. For any region on the loss manifold $\mathcal{S}$, the following holds:

**Theorem 4.1** (Liu et al. (2022)). *$\mathcal{L}(\theta)$ is $\mu$-PL if $\lambda_1(K_\theta) \geq \mu$ for all $\theta \in \mathcal{S}$.*

It follows that the learning behavior of transformers depends on the minimum eigenvalue of their tangent kernel. This suggests a connection between ORC and the training of transformers, given the link between ORC and the eigenvalue spectrum. We specify this connection next.

**Ricci curvature and convergence of gradient descent.** Given the tangent kernel of the transformer, $\varphi := (\nabla F)(\nabla F)^T$, from equations 5 and 6, we can compute the eigenvalues of $\varphi$. Thus, using the findings from (Dovonon et al., 2024) on $\lambda^A{}_i$ and $\lambda^H{}_j$ and in light of the implications of Theorem 4.1, under a set of standard assumptions (Appendix A), we prove the following lemma:

**Lemma 4.2.** *Let $p$ denote the probability of exponential convergence of gradient descent on the transformer. Then, $\frac{\partial p}{\partial \lambda^A_1} \geq 0$, i.e., the minimum attention eigenvalue is positively correlated with $p$.*

Meanwhile, monotonic functions of the minimum ORC bound the spectrum of the normalized Laplacian Bauer et al. (2011) (see Appendix A). Since the attention scores are outputs of the softmax activation, as we describe in Appendix A, there is a simple connection between the eigenvalues of $A$ and its Laplacian, which in turn implies that the spectrum of $A$ is also bounded by monotonic functions of the minimum ORC. It then follows from Lemma 4.2 that $p$ can be adjusted through shifting the distribution of ORC values of the attention matrix. This suggests that the ORC distribution bears implications for training transformers via gradient descent-based algorithms. Next, we propose a novel training heuristic to manipulate the ORC distribution in transformers, and we empirically validate this theoretical argument.

### 4.2 Training curvature-adjusted transformers

Section 3 discusses the novel connections we establish between attention maps curvature and various aspects of transformer behavior. This raises an intriguing possibility: manipulating the curvature distribution of the attention graph could allow us to adjust their behavior and improve their capabilities. A well-established tool to diffuse the curvature over a manifold is the Ricci flow (Hamilton, 1988), and its discrete variation could be applied to flatten the graph and reduce the frequency of edges with highly negative curvature (representing unstable networks) (Jin et al., 2008; Ni et al., 2019). However, each iteration of the discrete Ricci flow requires computing the curvature distribution, and hence it is computationally taxing to incorporate in the training. To overcome this challenge via a computationally efficient proxy, we introduce a simple, efficient, and effective method to manipulate the ORC distribution of attention maps through regularization by a function of the variance of attention scores. Aiming to either promote or reduce the frequency of low-curvature edges, this regularization strategy not only provides a means to empirically validate our theory, but also offers a practical mechanism to enhance the training and robustness of transformers by adjusting its curvature. The experiments in this section are performed using *L-ViT* and BERT-Tiny, described in Appendix D.

**Encouraging or discouraging low-curvature edges via regularization** Recognizing the correspondence between the weight distributions and ORC in a complete graph (of full attention), we propose a regularization method to control the attention variance to adjust the ORC distribution. Negative ORC corresponds to weight imbalance, while a weight distribution with a small variance corresponds to a positively-curved graph. Hence, we expect variance increase to promote smaller ORC values, while variance decrease will lead to their elimination. In Appendix C we elaborate on how the formulation of the ORC implies its connection with weight variance. To put this in practice, we regularize the loss function with a variance-dependent term $h(\sigma^2)$, such that $\frac{\partial h}{\partial \sigma^2} < 0$ for promoting low-curvature edges and $\frac{\partial h}{\partial \sigma^2} > 0$ for reducing their frequency. The total loss is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \gamma h(\sigma_A^2), \tag{7}$$

where $\mathcal{L}_{\text{task}}$ denotes the usual loss associated with the task (e.g., classification loss in our experiments), $\gamma$ the regularization coefficient, and $\sigma_A^2$ the variance of attention scores averaged over all layers. In our implementation, we use $h(x) = \exp(-ax + b)$ and $h(x) = \log(ax+b)$ to increase and decrease variance (respectively), with $a > 0$ and $b$ serving as tunable scale and shift parameters, helping to place the regularization value in an impactful range. Furthermore, the choice of exponential and logarithmic functions ensures decreasing marginals, limiting adjustments to the loss and avoiding extreme outcomes where attention either becomes uniformly distributed (losing discriminative power) or overly imbalanced (leading to instability or absurd attention).

Figure 4: Final layer ORC distributions in *L-ViT* trained via regularization methods increasing (↑) and decreasing (↓) variance, exhibiting a connection with the direction of variance regularization.

**Empirical results on curvature-adjusted training.** Our experiments on standard image and text datasets using *L-ViT* and BERT-Tiny confirm that our proposed regularization influences the ORC